FAQ Chatbot

Project Report - Task 2

1. Objective

Build a chatbot that answers customer questions for an online store (ShopEasy) by finding the most similar FAQ to the user's question using NLP and returning its answer in a chat interface.

2. Requirements and how they are met

Task requirement	Implementation
Collect FAQs	16 question and answer pairs on orders, delivery, payments and returns, stored in faqs.json
Preprocess text with NLP libraries	Lowercasing, tokenization (NLTK RegexpTokenizer), stopword removal, synonym mapping, Porter stemming (NLTK)
Match user question to most similar FAQ	TF-IDF vectors (scikit-learn) with cosine similarity
Display best matching answer	Best FAQ answer returned with the matched question and a confidence score; fallback reply below threshold 0.2
Optional: simple chat UI	Flask web app with chat bubbles, suggestion chips and a terminal mode

3. Technologies used

- Python 3, scikit-learn (TfidfVectorizer, cosine_similarity), NLTK (tokenizer and stemmer)
- Flask for the web server, HTML, CSS and JavaScript for the chat interface

4. How it works

- **Training:** each FAQ (question plus answer) is preprocessed and converted to a TF-IDF vector (unigrams and bigrams).
- **Preprocessing:** 'Where is my parcel?' becomes tokens such as 'track' and 'order' after stopword removal and synonym mapping.
- **Matching:** the user's question is vectorized the same way and compared with every FAQ by cosine similarity; the highest score wins.
- **Response:** if the best score is at least 0.2 the FAQ answer is shown; otherwise the bot asks the user to contact support.
- Greetings and thanks are handled with simple rules.

5. Sample results

User question	Matched FAQ
where is my parcel	How can I track my order?
i want my money back	How do I get a refund?
can i pay with upi	Which payment methods do you accept?

User question	Matched FAQ
do you ship abroad	Do you deliver internationally?
received broken item	What if I receive a damaged or wrong product?
what is the weather	No match - fallback message

6. How to run

- Extract faq_chatbot.zip and run `pip install -r requirements.txt`.
- Run `python app.py` and open http://127.0.0.1:5000, or run `python chatbot.py` for the terminal version.
- To change the topic, edit faqs.json.

7. Limitations and future scope

- TF-IDF matches words, not meaning; unusual phrasing may fail unless covered by the synonym list.
- Upgrade options: sentence embeddings (spaCy vectors or sentence-transformers), spell correction, conversation memory, a larger FAQ set.

8. FAQ data (faqs.json)

Question	Answer
How do I create an account?	Click 'Sign Up' at the top right, enter your name, email and a password, then confirm the verification email we send you.
I forgot my password. How can I reset it?	Click 'Forgot password' on the login page, enter your registered email and follow the reset link we send you. The link is valid for 30 minutes.
How can I track my order?	Go to 'My Orders' after logging in and click 'Track' next to your order. You will also receive tracking updates by email and SMS.
How long does delivery take?	Standard delivery takes 3-5 business days. Express delivery takes 1-2 business days in most cities.
What are the shipping charges?	Shipping is free on orders above Rs. 499. For smaller orders, a flat fee of Rs. 49 applies.
What is your return policy?	You can return most items within 7 days of delivery if they are unused and in original packaging.
How do I get a refund?	Once we receive and inspect your returned item, the refund is issued to your original payment method within 5-7 business days.
How can I cancel my order?	Open 'My Orders', select the order and click 'Cancel'. Orders can be cancelled until they are shipped.
Which payment methods do you accept?	We accept credit and debit cards, UPI, net banking, popular wallets and cash on delivery.
Is cash on delivery available?	Yes, cash on delivery is available for orders up to Rs. 10,000 in serviceable pin codes.
My payment failed but money was deducted. What should I do?	Do not worry. The amount is automatically refunded by your bank within 5-7 business days. If not, contact support with your transaction ID.
How do I apply a coupon code?	Enter the code in the 'Apply Coupon' box on the cart page before checkout and click 'Apply'.
Do you deliver internationally?	Currently we deliver only within India. International shipping will be added soon.
How can I contact customer support?	Email support@shopeasy.example or call 1800-000-000 between 9 AM and 9 PM, Monday to Saturday. You can also use this chat.
Can I change my delivery address after placing an order?	Yes, if the order has not shipped yet. Go to 'My Orders', select the order and click 'Edit address'.
What if I receive a damaged or wrong product?	Report it within 48 hours from 'My Orders' with a photo. We will arrange a free pickup and send a replacement or refund.

9. Source code: chatbot.py

```
"""FAQ chatbot core: collect FAQs -> preprocess -> TF-IDF + cosine similarity -> best answer."""
import json, re
from pathlib import Path
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# ---------- NLP preprocessing (NLTK if installed, simple fallback otherwise) ----------
try:
    from nltk.tokenize import RegexpTokenizer
    from nltk.stem import PorterStemmer
    _tok = RegexpTokenizer(r"[a-z0-9]+").tokenize
    _stem = PorterStemmer().stem
    NLTK = True
except ImportError:
    _tok = lambda t: re.findall(r"[a-z0-9]+", t)
    _stem = lambda w: re.sub(r"(ing|ed|es|s|ly)$", "", w) if len(w) > 4 else w
    NLTK = False

STOPWORDS =
    {"a","an","the","is","are","am","was","were","be","do","does","did","i","me","my","we","you","your",
 "it","its","to","of","in","on","at","for","and","or","can","could","would","should","how","what",

    "which","who","will","with","if","by","this","that","there","about","please","get","have","has","had","a
    ny",
 "from","as","so","but","not","no","yes"}
try:
    from nltk.corpus import stopwords
    STOPWORDS |= set(stopwords.words("english")) - {"not", "no"}
except Exception:
    pass

# Maps everyday wording to the words used in the FAQs (simple intent matching).
SYNONYMS =
    {"when":"long","parcel":"order","package":"order","purchase":"order","shipment":"order","where":"track",
 "pay":"payment","paying":"payment","upi":"payment","card":"payment","money":"refund","reimbursement":"refun
    d",
 "ship":"delivery","shipping":"delivery","deliver":"delivery","arrive":"delivery",
 "abroad":"international","overseas":"international","foreign":"international","outside":"international",
 "broken":"damaged","defective":"damaged","faulty":"damaged","wrong":"damaged",
 "discount":"coupon","promo":"coupon","voucher":"coupon","helpline":"support","help":"support",
 "signup":"account","register":"account","login":"account","send back":"return","exchange":"return"}

def preprocess(text: str) -> str:
    """lowercase -> tokenize (drops punctuation) -> remove stopwords -> map synonyms -> stem."""
    tokens = _tok(text.lower())
    tokens = [SYNONYMS.get(t, t) for t in tokens if t not in STOPWORDS]
    return " ".join(_stem(t) for t in tokens)

# ---------- Chatbot ----------
GREETINGS = {"hi","hello","hey","hii","good morning","good evening"}
THANKS = {"thanks","thank you","thx","ok thanks"}

class FAQBot:
    def __init__(self, path=None, threshold=0.2):
        path = Path(path or Path(__file__).with_name("faqs.json"))
        self.faqs = json.loads(path.read_text(encoding="utf-8"))
        self.threshold = threshold
        self.vec = TfidfVectorizer(ngram_range=(1, 2))
        self.matrix = self.vec.fit_transform([preprocess(f["q"] + " " + f["a"]) for f in self.faqs])

    def ask(self, text: str) -> dict:
        clean = text.strip().lower()
        if clean in GREETINGS:
            return {"answer": "Hello! Ask me anything about orders, delivery, payments or returns.",
    "score": 1.0}
        if clean in THANKS:
            return {"answer": "You're welcome! Anything else I can help with?", "score": 1.0}
        processed = preprocess(text)
        if not processed:
            return {"answer": "Please type your question.", "score": 0.0}
        scores = cosine_similarity(self.vec.transform([processed]), self.matrix)[0]
        best = int(scores.argmax())
        if scores[best] < self.threshold:
            return {"answer": "Sorry, I couldn't find an answer to that. Please contact
    support@shopeasy.example.",
                    "score": float(scores[best])}
        return {"answer": self.faqs[best]["a"], "matched": self.faqs[best]["q"], "score":
    float(scores[best])}

if __name__ == "__main__":
    bot = FAQBot()
```

```
    print("ShopEasy FAQ Bot (type 'quit' to exit)")
    while True:
        q = input("You: ")
        if q.strip().lower() in {"quit", "exit"}:
            break
        r = bot.ask(q)
        print(f"Bot: {r['answer']}  [score {r['score']:.2f}]")
```

10. Source code: app.py

```
from flask import Flask, request, jsonify, render_template
from chatbot import FAQBot

app = Flask(__name__)
bot = FAQBot()

@app.route("/")
def index():
    return render_template("index.html")

@app.route("/ask", methods=["POST"])
def ask():
    text = (request.get_json(silent=True) or {}).get("message", "")
    return jsonify(bot.ask(text))

if __name__ == "__main__":
    app.run(debug=True)
```

11. Source code: templates/index.html

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-
    scale=1">
<title>ShopEasy FAQ Chatbot</title>
<style>
:root{--bg:#f3f1ec;--card:#fff;--ink:#1c2321;--muted:#6a7571;--accent:#1f6f5c;--user:#1f6f5c;--bot:#e8efec}
@media(prefers-color-scheme:dark){:root{--bg:#121816;--card:#1a2420;--ink:#e8f0ed;--muted:#8fa39b;--
    accent:#4fc2a5;--user:#2a7d69;--bot:#25332e}}
*{box-sizing:border-box}body{margin:0;background:var(--bg);color:var(--ink);font-family:"Segoe UI",system-
    ui,sans-serif;display:flex;justify-content:center;height:100vh}
.chat{width:100%;max-width:560px;display:flex;flex-direction:column;background:var(--card)}
header{padding:16px 18px;border-bottom:1px solid var(--bot)}header h1{margin:0;font-size:1.1rem}header
    p{margin:2px 0 0;color:var(--muted);font-size:.85rem}
#log{flex:1;overflow-y:auto;padding:16px;display:flex;flex-direction:column;gap:10px}
.m{max-width:82%;padding:10px 13px;border-radius:14px;line-height:1.45;white-space:pre-wrap}
.bot{background:var(--bot);align-self:flex-start;border-bottom-left-radius:4px}
.user{background:var(--user);color:#fff;align-self:flex-end;border-bottom-right-radius:4px}
.hint{display:block;margin-top:6px;font-size:.75rem;color:var(--muted)}
.chips{display:flex;flex-wrap:wrap;gap:6px;padding:0 16px 8px}
.chips button{border:1px solid var(--accent);background:none;color:var(--accent);border-
    radius:99px;padding:5px 11px;font:inherit;font-size:.82rem;cursor:pointer}
form{display:flex;gap:8px;padding:12px;border-top:1px solid var(--bot)}
input{flex:1;padding:11px 13px;font:inherit;border:1px solid var(--bot);border-
    radius:10px;background:var(--bg);color:var(--ink)}
form button{background:var(--accent);color:#fff;border:0;border-radius:10px;padding:0
    18px;font:inherit;font-weight:600;cursor:pointer}
:focus-visible{outline:3px solid var(--accent);outline-offset:2px}
</style></head>
<body><div class="chat">
<header><h1>ShopEasy Help</h1><p>Ask about orders, delivery, payments and returns</p></header>
<div id="log" aria-live="polite"></div>
<div class="chips" id="chips"></div>
<form id="f"><input id="t" autocomplete="off" placeholder="Type your question" aria-label="Your
    question"><button>Send</button></form>
</div>
<script>
const log=document.getElementById("log"),t=document.getElementById("t");
function add(text,who,hint){const d=document.createElement("div");d.className="m "+who;d.textContent=text;
 if(hint){const s=document.createElement("span");s.className="hint";s.textContent=hint;d.appendChild(s)}
 log.appendChild(d);log.scrollTop=log.scrollHeight}
async function send(msg){if(!msg.trim())return;add(msg,"user");t.value="";
 try{const r=await fetch("/ask",{method:"POST",headers:{"Content-
    Type":"application/json"},body:JSON.stringify({message:msg})});
  const d=await r.json();add(d.answer,"bot",d.matched?"Matched FAQ: "+d.matched+"
    ("+Math.round(d.score*100)+"%)":"")}
 catch(e){add("Cannot reach the server. Is app.py running?","bot")}}
document.getElementById("f").onsubmit=e=>{e.preventDefault();send(t.value)};
["How can I track my order?","What is the return policy?","Which payments do you accept?"].forEach(q=>{
 const b=document.createElement("button");b.textContent=q;b.type="button";b.onclick=()=>send(q);document.get
    ElementById("chips").appendChild(b)});
add("Hi! I'm the ShopEasy assistant. How can I help you today?","bot");
</script></body></html>
```